Ardeola 54(2), 2007, 331-334

AN IMPROVED METHOD OF EXTRACTING DEGRADED DNA SAMPLES FROM BIRDS AND OTHER SPECIES

UN MÉTODO GENERAL PARA LA EXTRACCIÓN DE ADN DEGRADADO EN MUESTRAS DE AVES Y OTRAS ESPECIES

Fernando ALDA* [1], Isabel REY** and Ignacio DOADRIO*

In recent years, non-invasive genetic sampling (NGS) has proved to be a powerful tool in population and conservation genetics (Khon and Wayne, 1997; Taberlet *et al.,* 1999). It is often applied to elusive or endangered species because NG samples are easy to find in the field and do not require interaction with the animals. Analysis of this data permits species identification, sexing, tracking of free-ranging animals, population census, parental analysis... (Reed *et al.,* 1997; Segelbacher and Steinbrück, 2001; Palsbøll *et al.,* 1997; Khon *et al.,* 1999; Field *et al.,* 1998).

DNA can be isolated from basically any biological sample found in the field and later amplified by PCR. Another important source of DNA are museum samples (skins, bones, etc), which can provide important molecular data about historical populations. Collecting all this available information depends on being able to isolate the highest quality DNA in the greatest quantities.

Most of the problems that could affect the reliability of NGS techniques are related to the PCR (Taberlet *et al.,* 1996; Gagneux *et al.,* 1997). Difficulty obtaining PCR products, allele dropout and false alleles (Taberlet *et al.,* 1996) are either directly or indirectly related to DNA extraction. A dirty extraction product will contain excessive PCR inhibitors; a low extraction yield will not give enough templates for amplification or can increase the risk of allele dropout. Overcoming these problems is crucial especially for individualizations based on a single sample, where genotyping errors would provide an erroneous result (Waits *et al.,* 2001).

DNA extraction depends first on the sample type, (*e.g.*, hairs and feathers contain better DNA quantity and quality than faeces), then length of time the sample is exposed to the environment before collection (*e.g.*, samples that have been in the field for a long periods of time, or that have been exposed to humidity, radiation etc. would have more degraded DNA), collection and preservation method (dried, ethanol, frozen...; Frantzen *et al.,* 1998), and the time elapsed between collection and extraction (Roon *et al.,* 2003; Nsubuga *et al.,* 2004). Finally the extraction protocol (phenol chloroform, Chelex ®, silica-Guanidine Thiocyanate, diatomaceous earth, QIAmp DNA stool mini kit, magnetic beads,…) has to be suitable for

* Departamento de Biodiversidad y Biología Evolutiva, Museo Nacional de Ciencias Naturales (CSIC), calle José Gutiérrez Abascal 2, Madrid, E-28006 Spain.

** Colección de Tejidos y ADN, Museo Nacional de Ciencias Naturales (CSIC), calle José Gutiérrez Abascal 2, Madrid, E-28006 Spain.

1 Corresponding author: alda@mncn.csic.es

the studied samples (Vigilant, 1999; Pichler *et al.,* 2001; Parsons *et al.,* 1999; Segelbacher and Steinbrück, 2001; Flagstad *et al.,* 1999).

In this study, the results obtained were not sufficiently adequate using traditional methods (Boom *et al.,* 1990; Walsh *et al.,* 1991) or kits (QIAmp DNA stool mini kit, QIAGEN), therefore it was decided to develop a protocol based on other fossil DNA methods (Yang *et al.,* 1998), that had simple steps, did not require major changes to the protocol depending on the sample, and that could be used for a wide range of sample types.

The performance of the new protocol was tested with 79 capercaillie *Tetrao urogallus*, faeces collected in the field during the months of April, May and June of 2003. DNA was extracted three times from all the faecal samples using a silica gel and guanidine thiocyanate protocol (Boom *et al.,* 1990), and extracted one time following the new protocol. Silica gel was chosen because it was necessary to preserve the DNA long enough to perform all the repeats. Other methods such as Chelex degrade the DNA when stored for long time periods (Walsh *et al.,* 1991) and no posterior comparisons of genotypes are possible. The usefulness of the method was also evaluated in different sample types from capercaillie and other species (*e.g.*, brown bear *Ursus arctos* and Iberian lynx *Lynx pardinus*).

A small amount of faecal sample was crumbled, or surface washed with PBS, in 15 ml Falcon tubes, and incubated in 2 ml of extraction buffer (0.5M EDTA pH 8.0, 0.5 % SDS, and 50 µg/ml Proteinase K) overnight at 55ºC in a shaking water bath. The tubes were centrifuged for 10' at 2000 g. Supernatant was recovered. To this, we added the same volume of 4.5M Guanidine thiocyanate, 0.5M Potassium Acetate pH 5.0 buffer. One QIAquick column was placed for each sample. Seven hundred and fifty microliters of the sample was loaded and spun for 1 minute at 12800 g. Flowthrough was discarded and the process was repeated until all the sample liquid had been passed through the column. Two washes of 450µl and 700µl PE buffer were performed. The DNA was eluted in100 µl of TE buffer after 10' incubation at room temperature and centrifuged 5' at 12800 g. Extractions were made in a room exclusively dedicated to ancient DNA processing, and included a negative control to check for exogenous contamination.

For feather and hair extraction, 2 - 5mm of the tip (depending on the feather size) or of the hair root was cut longitudinally and transversally and incubated in 1.5 ml of the above-mentioned extraction buffer. The same protocol previously described was followed except only one 750 µl PE buffer wash was needed.

The results of the extractions were tested by PCR amplification of a species specific microsatellite locus (Segelbacher *et al.,* 2000). Amplification was carried out in 10µl reaction volume (1x GeneAmp Buffer II, 2mM $MgCl_2$, 0.1µM of each primer, 0.2 mM dNTP, 0.1 µg/ul BSA, 0.01U of TaqGold Taq polymerase (Applied Biosystems) and 1µl of the extraction product. PCRs consisted of 10' of initial denaturation at 95ºC, followed by 50 cycles of 30" at 95ºC, 45" at 55ºC and 45" at 72ºC, concluding with a final extension step of 5' at 72ºC. All PCR reactions included a negative and a positive control. When a band of the expected size was observed in a 1.5 % agarose gel, the extraction was considered "successful". If no band was observed, the extraction was considered to be "unsuccessful".

In the event that an extraction was unsuccessful or the elution was dirty, it was passed through a new QIAquick column following the manufacturer's protocol and eluted in 100µl of TE. The columns were saved in case the extraction results were negative again, so a third clean-up of the product could be performed with the same column.

The three independent silica gel extractions yielded 14, 19 and 18 successful extractions (17.7 %, 24.1 % and 22.8 %, respectively). Out of 237 extraction runs (79 samples x 3), a total of 27 unique successful DNA extractions

(34.2 %) was obtained. For the QIAmp column-based protocol, 2 samples gave a positive PCR result right after extraction. After being passed a second time through a new column, 50 (63.3 %) samples were successfully amplified. Samples that still were negative were passed through the same column a third time and 11 of them were amplified. Sixty-three samples (79.7 %) were extracted successfully with this method.

Variations of other extraction methods were also tried, such as two consecutive silica gel extractions, and purification of the extraction product with a QIAmp column. None of them produced significantly better results than the single silica gel extraction. At first, a fourth pass through the columns was performed but it did not yield additional successful samples.

Feathers, hairs and museum skins produced an 85-95% of successful extractions. These results surpassed those obtained in similar studies (Segelbacher, 2002) that produced 50 % positive genotypes from moulted feathers.

The new method shortens the incubation periods of Yang's protocol (1998) and significantly reduces the sample volumes that must be filtered through the column, thanks to the use of different buffers. More important is the possibility of extracting DNA, with one single and easy method, from a wide range of samples of different kind and species. Thus, it was possible to isolate DNA from; faeces, shed hairs and teeth of brown bears; faeces, shed hairs and musem skins of Iberian lynx; and faeces, moulted feathers, museum skins and egg shells of capercaillie.

In conclusion, this protocol seems to be a more efficient method for NGS, saving both time and financial resources in the laboratory and in the field.

RESUMEN.—*El uso de muestras no invasivas se ha convertido una técnica ampliamente utilizada en estudios de genética y conservación. La mejora del método de extracción de ADN es el primer paso para evitar errores en el genotipado. En este trabajo se describe un método de extracción que únicamente requiere que el ADN sea incubado durante una noche y posteriormente purificado con columnas QIAquick (QIAGEN). Se extrajeron 79 excrementos de urogallo común* Tetrao urogallus *empleando tanto el nuevo método como un método tradicional de sílice y guanidín tiocianato. Se comprobó que el nuevo método es más eficiente y adecuado para diferentes tipos de muestras: excremento de ave y mamífero, plumas mudadas, pelos, dientes y pieles de museo.*

ACKNOWLEDGEMENTS.—We thank everybody who provided samples: Rangers of Principado de Asturias, José Luis Benito (ESGERENA) and Fundación CBD-Habitat. Thanks also to José Luis García and Lourdes Alcaraz for their assistance in the laboratory and useful comments and to James Watkins who corrected the English text. This work is part of a research project for the genetic characterization of the Cantabric capercaillie supported by the Principado de Asturias. F.A. benefited from an I3P-CSIC grant.

BIBLIOGRAPHY

BOOM, R., SOL, C. J. A., SALIMANS, M. M. M., JANSEN, C. L., WERTHEIM-VAN DILLEN, P. M. E. and VAN DER NOORDAA, J. 1990. Rapid and simple methods for purification of nucleic acids. *Journal of Clinical Microbiology*, 28: 495-503.

FIELD, D., CHEMNICK, L., ROBBINS, M., GARNER, K. and RYDER, O. 1998. Paternity determination in captive lowland gorillas and orangutans and wild mountain gorillas by microsatellite analysis. *Primates*, 39: 199-209.

FLAGSTAD, Ø., RØED, K., STACEY, J. E. and JAKOBSEN, K. S. 1999. Reliable noninvasive genotyping based on excremental PCR of nuclear DNA purified with a magnetic bead protocol. *Molecular Ecology*, 8: 879-883.

FRANTZEN, M. A., SILK, J. B., FERGUSON, J. W., WAYNE, R. K. and KOHN, M. H. 1998. Empirical evaluation of preservation methods for faecal DNA. *Molecular Ecology*, 7: 1423-1428.

GAGNEUX, P., BOESCH, C. and WOODRUFF, D. S. 1997. Microsatellite scoring errors associated with non-invasive genotyping based on nuclear DNA amplified from shed hair. *Molecular Ecology*, 6: 861-868.

JOHNSON, W. E., GODOY, J. A., PALOMARES, F., DELIBES, M., FERNANDES, M., REVILLA, E., and O'BRIEN, S. J. 2004. Phylogenetic and phylogeographic analysis of Iberian Lynx populations. *Journal of Heredity*, 95: 19-28.

KOHN, M. H. and WAYNE, R. K. 1997. Facts from faeces revisited. *TREE*, 12**:** 223-227.

KOHN, M., YORK, E., KAMRADT, D., HAUGHT, G., SAUVAJOT, R. and WAYNE, R. 1999. Estimating population size by genotyping faeces. *Proceedings of the Royal Society of London Series B*, 266: 657-663.

NSUBUGA, A. M., ROBBINS, M. M., ROEDOR, A. D., MORIN, P. A., BOESCH, C. and VIGILANT, L. 2004. Factors affecting the amount of genomic DNA extracted from ape faeces and the identification of an improved sample storage method. *Molecular Ecology,* 13: 2089-2094.

PALSBØLL, P. J., ALLEN, J., BÉRUBÉ M, CLAPHAM, P. J., FEDDERSEN, T. P., HAMMOND, P. S. HUDSON, R. R., JORGENSEN H., KATONA, S., LARSEN, A. H., LARSEN, F., LIEN, J., MATTILA, D. K., SIGURJONSSON, J., SEARS, R., SMITH, J., SPONER, R., STEVICK, P., and OIEN, N. 1997 Genetic tagging of humpback whales. *Nature*, 388: 767-769.

PARSONS, K. M., DALLAS, J. F., CLARIDGE, D. E., DURBAN, J. W., BALCOMB III, K. C., THOMPSON, P. M. and NOBLE, L. R. 1999 Amplifying dolphin mitochondrial DNA from faecal plumes. *Molecular Ecology*, 8: 1766-1768.

PICHLER, F. B., DALEBOUT, M. L. and BAKER, C. S. 2001. Nondestructive DNA extraction from sperm whale teeth and scrimshaw. *Molecular Ecology Notes*, 1: 106-109.

REED, J. Z., TOLLIT, D. J., THOMPSON, P. M. and AMOS, W. 1997. Molecular scatology: the use of molecular genetic analysis to assign species, sex and individual identity to seal species. *Molecular Ecology*, 6: 225-234.

ROON, D. A., WAITS, L. P. and KENDALL, K. C. 2003. A quantitative evaluation of two methods or preserving hair samples. *Molecular Ecology Notes*, 3: 163-166.

SEGELBACHER, G., PAXTON, R., STEINBRÜK, G., TRONTELI, P. and STORCH, I. 2000. Characterization of microsatellites in capercaillie *Tetrao urogallus. Molecular Ecology*, 9: 1919-1952.

SEGELBACHER, G. and STEINBRÜK, G. 2001. Bird faeces for sex identification and microsatellite analysis. *Die Vogelwarte*, 41: 139-142.

SEGELBACHER, G. 2002. Noninvasive genetic analysis in birds: testing reliability of feather samples. *Molecular Ecology Notes*, 2: 367-369.

TABERLET, P., GRIFFIN, S., GOOSSENS B., QUESTIAU, S., MANCEAU, V., ESCARAVAGE, N., WAITS, L. P. and BOUVET, J. 1996. Reliable genotyping of samples with very low DNA quantities using PCR. *Nucleic Acids Research*, 24: 3189–3194.

TABERLET, P., WAITS, L. P. and LUIKART, G. 1999. Noninvasive genetic sampling: look before you leap. *TREE*, 14: 323–327.

VIGILANT, L. 1999. An evaluation of techniques for the extraction and amplification of DNA from naturally shed hairs. *Biological Chemistry*, 380: 1329-1331.

WAITS, L. P., LUIKART, G. and TABERLET, P. 2001. Estimating the probability of identity among genotypes in natural populations: cautions and guidelines. *Molecular Ecology*, 19: 249-259.

WALSH, P., METZGER, D. and HIGUCHI, R. 1991. Chelex 100 as a medium for simple extraction of DNA for PCR-based typing from forensic material. *Biotechniques*, 10: 506-513.

YANG, D. Y., ENG, B., WAYE, J. S., DUDAR, J. C. and SAUNDERS, S. R. 1998. Technical note: improved DNA extraction from ancient bones using silica-based spin columns. *American Journal of Physical Antrophology*, 105: 539-543.

[Recibido: 11-04-07]
[Aceptado: 27-09-07]